How-To Library

Informatica

Address Verification Best Practices for Finland

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Abstract

This document tells you how to connect the elements in a Finland address to the appropriate address elements in Informatica Address Verification. This document also describes best practices that you can implement when you verify addresses in Finland.

Supported Versions

- Informatica Address Verification 3.0 and later
- Informatica Address Verification (On-Premises) 5.x

Table of Contents

Introduction. . . . 2
Process Modes. . . . 2
Character Set. . . . 3
Address Format and Address Structure. . . . 3
Key Address Elements. . . . 3
Mapping Input Address Data to Address Elements. . . . 3
Input and Result XML in Address Verification (On-Premises). . . . 4
Input and Result XML in Address Verification. . . . 5
Address Enrichments. . . . 6
Address Verification Tips and Best Practices. . . . 6
Frequently Asked Questions. . . . 7
Reference Address Databases. . . . 8
Additional Documentation Resources. . . . 8

Introduction

Informatica Address Verification can validate and enhance addresses in Finland. This how-to article provides you with configuration information and best practices that you can follow to get the most from the addresses. The best practices include advice on how to select the appropriate XML elements for Finland address data.

To verify the quality of an input address, Address Verification compares the address information to the data in one or more reference data files. You can configure Address Verification to identify the different types of address information in the input data. You must map each item of address information to an appropriate element in Address Verification.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies determines the elements that you need.

Process Modes

Informatica Address Verification supports the following process modes for Finland addresses:

- BATCH
- COUNTRY RECOGNITION
- FAST COMPLETION

- INTERACTIVE
- PARSE

Character Set

Finland reference address databases store address data in the Latin-1 and ASCII scripts. By default, Informatica Address Verification returns the verified Finland addresses in the Latin-1 script.

The Latin-1 script includes the characters å, ä, and ö. Latin-1 does not contain the characters š and ž, which occur occasionally in loan-words in Finnish.

Address Format and Address Structure

You improve the performance of Address Verification when you define the input address in the format that the postal service expects.

The Finland postal service recognizes the following structure for a postal address:

```
Line 1: RECIPIENT OR CONTACT NAME
Line 2: STREET NAME HOUSE NUMBER [SUB-BUILDING INFORMATION]
Line 3: [FI-] POSTAL CODE CITY
Line 4: COUNTRY
```

The following sample address illustrates the structure:

```
MATTI MEIKÄLÄINEN
KUUSIKKOTIE 1
00630 HELSINKI
FIN
```

Key Address Elements

Address Verification is unlikely to verify an address successfully if one of the following items is absent from the address:

- Locality
- Postal Code
- Street

Mapping Input Address Data to Address Elements

To verify an address, map the input address information to appropriate elements in Address Verification.

The following table shows how you can map the input address to the address elements:

Information Type	Address Information	Address Elements
Street Name	KUUSIKKOTIE	Street
House Number	1	HouseNumber
Postal Code	00630	PostalCode

Information Type	Address Information	Address Elements
City	HELSINKI	Locality
Country	FINLAND	Country

Input and Result XML in Address Verification (On-Premises)

The following example shows how you can map the address information to the XML address elements in Address Verification (On-Premises).

You enter the following input address:

```
<InputData>
    <AddressElements>
        <Country Item="1" Type="NAME">FINLAND</Country>
    </AddressElements>
    <AddressLines>
        <FormattedAddressLine Line="1">KUUSIKKOTIE 1</FormattedAddressLine>
        <FormattedAddressLine Line="2">00630 HELSINKI</FormattedAddressLine>
    </AddressLines>
</InputData>
```

Informatica Address Verification (On-Premises) returns the following result:

```
<?xml version="1.0" encoding="UTF-16"?>

<!-- EngineVersion="5.14.1.47633" -->

<Result     ProcessStatus="V4"
     ModeUsed="BATCH"
     Count="1"
     CountOverflow="NO"
     CountryISO3="FIN"
     PreferredScript="DATABASE"
     PreferredLanguage="DATABASE">

<ResultData ResultNumber="1"
     MailabilityScore="5"
     ResultPercentage="100.00"
     ElementResultStatus="F0F080F0F000000000E0"
     ElementInputStatus="60600060600000000060"
     AddressResolutionCode="00000000000000000000"
     ExtElementStatus="00500050000000000000"
     ElementRelevance="10100010100000000010"
     AddressType="S"
     LanguageISO3="FIN">
    <AddressElements>
        <Country Type="NAME_EN" Item="1">Finland</Country>
        <Locality Item="1">Helsinki</Locality>
        <PostalCode Item="1">00630</PostalCode>
        <Province Item="1">Helsinki-Uusimaa</Province>
        <Street Item="1">Kuusikkotie</Street>
        <Number Item="1">1</Number>
    </AddressElements>
    <AddressLines>
        <DeliveryAddressLine Line="1">Kuusikkotie 1</DeliveryAddressLine>
        <CountrySpecificLocalityLine Line="1">FI-00630 Helsinki</CountrySpecificLocalityLine>
        <FormattedAddressLine Line="1">Kuusikkotie 1</FormattedAddressLine>
        <FormattedAddressLine Line="2">FI-00630 Helsinki</FormattedAddressLine>
        <FormattedAddressLine Line="3">Finland</FormattedAddressLine>
    </AddressLines>
    <AddressComplete>Kuusikkotie 1
FI-00630 Helsinki
Finland</AddressComplete>
</ResultData>
</Result>
```

Input and Result XML in Address Verification

The following example shows how you can verify a Finland address in a SOAP call to Address Verification.

You submit the following request:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:v="http://
validator5.AddressDoctor.com/Webservice5/v4">
    <soap:Header/>
    <soap:Body>
        <v:Process>
            <v:login>****</v:login>
            <v:password>****</v:password>
            <v:parameters>
                <v:ProcessMode>BATCH</v:ProcessMode>
                <v:ServiceParameters></v:ServiceParameters>
                <v:ValidationParameters>
                </v:ValidationParameters>
            </v:parameters>
            <v:addresses>
                <v:Address>
                    <v:FormattedAddress>
                        <v:string>KUUSIKKOTIE 1</v:string>
                        <v:string>00630 HELSINKI</v:string>
                        <v:string>FINLAND</v:string>
                    </v:FormattedAddress>
                </v:Address>
            </v:addresses>
            <v:enrichments>
            </v:enrichments>
        </v:Process>
    </soap:Body>
</soap:Envelope>
```

Informatica Address Verification returns the following response:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
   <soap:Body>
      <ProcessResponse xmlns="http://validator5.AddressDoctor.com/Webservice5/v4">
         <ProcessResult>
            <StatusCode>100</StatusCode>
            <StatusMessage>OK</StatusMessage>
            <Results>
              <Result>
                  <ProcessStatus>V4</ProcessStatus>
                  <CountryISO3>FIN</CountryISO3>
                  <ResultDataSet>
                     <ResultData>
                        <ResultNumber>1</ResultNumber>
                        <MailabilityScore>5</MailabilityScore>
                        <ResultPercentage>100.00</ResultPercentage>
                        <ElementInputStatus>6060006060000000060</ElementInputStatus>
                        <ElementResultStatus>F0F080F0F000000000E0</ElementResultStatus>
                        <ElementRelevance>10100010100000000010</ElementRelevance>
                        <ExtElementStatus>0050005000000000000</ExtElementStatus>
                        <AddressResolutionCode>00000000000000000000</AddressResolutionCode>
                        <AddressType>S</AddressType>
                        <LanguageISO3>FIN</LanguageISO3>
                        <Address>
                           <Street>
                              <string>Kuusikkotie</string>
                           </Street>
                           <HouseNumber>
                              <string>1</string>
                           </HouseNumber>
                           <Locality>
                              <string>Helsinki</string>
                           </Locality>
                           <PostalCode>
                              <string>00630</string>
```

```
                              </PostalCode>
                              <Province>
                                 <string>Helsinki-Uusimaa</string>
                              </Province>
                              <Country>
                                 <string>FINLAND</string>
                              </Country>
                              <DeliveryAddressLines>
                                 <string>Kuusikkotie 1</string>
                              </DeliveryAddressLines>
                              <CountrySpecificLocalityLine>
                                 <string>FI-00630 Helsinki</string>
                              </CountrySpecificLocalityLine>
                              <FormattedAddress>
                                 <string>Kuusikkotie 1</string>
                                 <string>FI-00630 Helsinki</string>
                                 <string>FINLAND</string>
                              </FormattedAddress>
                              <AddressComplete>Kuusikkotie 1;FI-00630 Helsinki;FINLAND</
AddressComplete>
                           </Address>
                        </ResultData>
                     </ResultDataSet>
                  </Result>
               </Results>
            </ProcessResult>
         </ProcessResponse>
      </soap:Body>
   </soap:Envelope>
```

Address Enrichments

You can configure Informatica Address Verification to add enrichments to verified Finland addresses.

The following table lists the types of enrichment that you can add to a Finland address:

Enrichment Type	Description
CAMEO	You can retrieve CAMEO profiles as enrichments for valid Finland addresses. CAMEO profiles are consumer classification systems that provide socio-economic and demographic indicators about the residents at the address.
Standard Geocoding	You can retrieve standard, interpolated geocoordinates for valid Finland addresses.
Arrival Point Geocoding	You can retrieve arrival point geocoordinates for valid Finland addresses.

Address Verification Tips and Best Practices

Note the following points when you verify Finland addresses:

- Address Verification validates a Finland address up to house number level.
- For best results, enter individual house numbers in Finland addresses instead of the house number range.
- Address Verification is configured to return valid address information in the Latin-1 script by default. The `PreferredScript` attribute determines the character set in which Address Verification returns the address data.

The following table shows how Address Verification returns address data for different PreferredScript settings in Finland:

Country	DATABASE	POSTAL _ADMIN _PREF	POSTAL _ADMIN _ALT	LATIN	LATIN _ALT	LATIN _1
FIN	Latin-1	Latin-1	Latin-1	ASCII	Latin-1	Latin-1

Note: The Latin-1 character set contains additional characters that can appear in Finnish and Swedish, such as å, ä, and ö. ASCII does not contain the additional characters. If you choose the ASCII script as an output language, and if your input data includes the extended characters, Address Verification might fail on rare occasions to properly validate an input element that uses the characters. To mitigate against this issue, include delimiters for all address elements in the input address.

Frequently Asked Questions

Can Address Verification return address information in multiple languages for Finland?

Address Verification can return address information in the Finnish language or the Swedish language. The default language is Finnish. Address Verification can return all address information in the Finnish language and return street, locality, and province information in Swedish.

To set the preferred language, configure the `PreferredLanguage` attribute of the Result element in `Parameters.xml`.

The following table describes the `PreferredLanguage` values that you can set for Finland addresses:

Option	Description
DATABASE	Default value. Returns all address information in Finnish.
ALTERNATIVE_1	Returns all address information in Finnish.
ALTERNATIVE_2	Returns the street, locality, and province information in Swedish. Returns all other information in Finnish.
PRESERVE_INPUT	Returns the address information in the input language if the information is in the reference data. If Address Verification detects more than one supported language in the input address, it returns the address in the database language. If Address Verification cannot return an element in the input language, it returns the element in the database language.

Does the reference data for Finland contain sub-building data?

No. Finland reference data does not contain sub-building data.

Does Address Verification require the FI- prefix before the post code?

The FI- prefix is optional. Address Verification generates the same result for an address with or without the prefix. You can include the prefix on an unfielded address line, such as formatted address, country-specific locality line, or complete address. Address Verification returns the FI- prefix in unfielded address lines. Address Verification does not return the prefix within the PostalCode address element.

Reference Address Databases

Informatica updates the Finland reference address databases quarterly. You can download the latest database packages from the Informatica Database Download portal. For news about reference address databases and software updates for Address Verification, subscribe to the newsletter from Informatica.

Informatica Address Verification provides the following reference databases for Finland addresses:

- `FIN5BI.MD`. Batch and interactive data.
- `FIN5CA.MD`. CAMEO data.
- `FIN5FC.MD`. Fast completion data.
- `FIN5GC.MD`. Standard geocoding data.
- `FIN5GCAP.MD`. Arrival point geocoding data.

Note: You download databases for Informatica Address Verification (On-Premises). You do not need to download databases for Informatica Address Verification, as Informatica Address Verification runs on Informatica Cloud®.

Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, click the following links:

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor-cloud

Author

Shahani Natalia Mendonca

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